

ATIQRUR RAHAMAN

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Education

Bangladesh University of Engineering & Technology (BUET)

Dhaka, Bangladesh

B.Sc. in Naval Architecture and Marine Engineering

Jan. 2022 – Jun. 2026

- CGPA: 3.78 / 4.00 (3rd in the Department)

Research Interest

- CFD
- Heat Transfer
- Offshore Structure
- Phase Changing Material
- Battery Thermal Management

Research Experience

- **Undergraduate Thesis Student, NAME, BUET** (Overview) *Apr. 2025 – Jun. 2026*

Thesis title: *Numerical Investigation of Mooring Line Angular Spacing on the Dynamic Motion of a 15MW Semi-Submersible Offshore Wind Turbine Platform.*

Under the supervision of: Dr. Md. Shahidul Islam, NAME, BUET

- Modelled the IEA 15 MW reference turbine on the UMaine VolturnUS-S platform
- Coupled frequency-domain and time-domain analysis in **ANSYS AQWA**
- Swept mooring line spread angles across a range of wave headings

- **Independent Research Collaboration** (Overview) *Feb. 2025 – Present*

Project title: *Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio.*

A self-organized collaboration with friends and BUET alumni, outside any university-affiliated project or course.

- Studied K-shaped fins in a PCM enclosure in **ANSYS Fluent 2024R1**
- Swept a range of fin angles and arm length ratios; intermediate angles with a longer lower arm raised melting efficiency
- Contribution: conceptualization, simulations, formal analysis, data curation, and production of the contour plots, efficiency comparisons, and LMLI curves

- **Ongoing Research** (Overview) *Aug. 2026 – Present*

Project title: *Iso-Volume Comparison of Fin Geometries for PCM-Based Battery Thermal Management: Shape Ranking under Repeated Charge and Discharge Cycling.*

- Ranking fin geometries for PCM-based battery thermal management in **ANSYS Fluent**
- Measuring how long each arrangement holds the cell below its safe operating temperature
- Running many charge and discharge cycles, tracking residual melt fraction between cycles

- **International Student Design Competition for Safe and Affordable Ferry organized by Worldwide Ferry Safety Association (WFSA)** (Overview) *2025*

Project title: *Naija Spirit – a 200-passenger battery-electric catamaran for the Ikorodu–CMS route across the Lagos lagoon.*

Team Black Pearl, Department of Naval Architecture and Marine Engineering, BUET

- Designed the auxiliary propulsion system, including a hydrogen fuel cell arrangement, and calculated wood and outfit weight
- Assisted in sourcing propulsion system-related machinery and wrote the report for this portion of the design

- **13th International Student Design Competition for a Safe and Affordable Ferry (2025–26), organized by Worldwide Ferry Safety Association (WFSA)** [\(Overview\)](#) 2026

Project title: *MV Niger Drift – a 200-passenger, CNG-powered ferry for Nigeria’s Onitsha–Lokoja route, cruising at 20 knots with a travel time of about 5½ hours per trip.*

Team Black Pearl, Department of Naval Architecture and Marine Engineering, BUET

Faculty Advisors: Dr. Zobair Ibn Awal, Md. Abdul Kader

- Designed the hybrid propulsion system (CNG generators plus battery) and the CNG tank, station, and pontoon arrangement
- Selected and sized the thrusters, propulsion motors, solar panels, and hydrokinetic turbine
- Drew the machinery arrangement and electric circuit diagram in AutoCAD, and specified the navigation aids

- **Junior Year Design Project, BUET** [\(Overview\)](#) Jan. 2024 – Feb. 2025

Project title: *Design of a 175 Passenger Ferry for circular waterways around Dhaka.*

Under the supervision of: Dr. Mir Tareque Ali, NAME, BUET

- Worked through the design spiral from principal dimensions to the lines plan
- Calculated hydrostatics, resistance, and powering in linked Excel sheets
- Estimated weight and checked stability against the design requirements
- Selected the propeller and ran a cavitation check
- Sized the engine, gearbox, and engine foundation

Publications

- Dr. Md. Shahidul Islam, **Md. Atiqur Rahaman**, and Md Tanvir Anjum Zenith, “Effect of Mooring Line Angular Spacing on the Wave-Frequency Motion Response of a 15 MW Semi-Submersible Floating Offshore Wind Turbine Based on AQWA” *(In Preparation)*. [\(Overview\)](#)
- Md Tanvir Anjum Zenith, Adri Das, Shafollo Proshad Gupto, **Md. Atiqur Rahaman**, Md. Khairul Islam, “Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio.” *(Submitted for Review in Case Studies in Thermal Engineering.)* [\(Overview\)](#)

Professional Experience

Intern [\(Overview\)](#) Jan. 2024 – Feb. 2025

Dockyard & Engineering Works Ltd, BANGLADESH NAVY – Narayanganj, Bangladesh

- Completed a three-week industrial attachment at a working shipyard
- Trained on marine propulsion systems, shipboard air conditioning, thermal management of engine rooms, and CNC plate cutting

Technical Skills

Simulation Software	ANSYS Fluent, ANSYS AQWA, ABAQUS
Programming Languages	C++, Python, Fortran
CAD & 3D Modelling	AutoCAD, Rhinoceros 3D
Ship Design Software	MAXSURF, DELFTship
Referencing Software	Zotero
Soft Skills	Leadership, Team Work, Mass Communication

Honors and Awards

- **University Merit Scholarship** (8 times in 8 semesters)
- **University Dean’s List Scholarship** (2 times, in 1st and 2nd year)
- Education Board Technical Scholarship, BUET

- 1st prize in “NAME Olympiad” at NAME Fest 2023
- **1st Prize (Champion)** and **\$5,000 prize money**, Worldwide Ferry Safety Association Design Competition 2025, Team Black Pearl
- **2nd Runner-up** and **\$1,000 prize money**, Worldwide Ferry Safety Association Design Competition 2026, Team Black Pearl
- Award and certificate from the Chairman of BIWTC in recognition of outstanding performance in the Ferry Safety Association Design Competition

Extracurricular Activities

- University Class Representative, Department of Naval Architecture and Marine Engineering, BUET *Jan. 2022 – Feb. 2023*
- **Volunteer, MARTEC 2022** – International Conference on Marine Technology *Dec. 2022*
- Team Member, Black Pearl, BUET, Worldwide Ferry Safety Association Design Competition *Sept. 2024 – Present*
- Former Vice President, BUET Photographic Society *Jun. 2025 – Jul. 2026*
- Former Director of Design and Creation, BUET Entrepreneurship Development Club *Jun. 2025 – Jul. 2026*

References

Dr. Mir Tareque Ali	Dr. Md. Shahidul Islam
Professor, Dept. of Naval Architecture and Marine Engineering, BUET	Professor, Dept. of Naval Architecture and Marine Engineering, BUET
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